

performing well on any downstream NLP task are acquired by models during the ‘next word prediction’ pretraining task. For instance, models learn syntax and, to some extent, semantics as part of their pretraining.

The key advantage of using ‘next word prediction’ as the pretraining task is the fact that we don’t need any labelled data. The models can be trained in a self-supervised manner on any large text corpus available. There have been variations in ‘next word prediction’ tasks, which are typically unidirectional — from left to right. So in order to capture bidirectional context, language models are pretrained on masked token prediction, next sentence prediction, etc, which are again self-supervised and do not require explicit labelled data.

Large language models have extremely large-sized parameter space. GPT-3 has 175 billion parameters, and the recently introduced Megatron-Turing NLG has 530 billion parameters (three times that of GPT-3). China recently announced an LLM called Wu Dao that has 1.75 trillion parameters, which is 10 times that of GPT-3. While ‘large language models’ (LLM) are a subset of ‘pretrained models’ (PTM), you will often find these terms being used interchangeably in the literature. For example, CLIP from openAI is a pretrained model for text-driven image retrieval tasks. Similarly, Resnet-50 is a pretrained model (trained on ImageNet data) for computer vision.

So what is the definition of a ‘foundation model’? According to the Stanford AI report on foundation models, “A foundation model is any model that is trained on broad data at scale and can be adapted (e.g., fine-tuned) to a wide range of downstream tasks.” As that paper points out, the two key characteristics of ‘foundation models’ are homogenisation of models and emergence of new functionalities. In this context, emergence indicates that certain behaviours of the machine learning system are implicitly induced rather than explicitly constructed or learnt. For example, LLMs are good at syntax even though they have not explicitly taught grammar rules as a task. While deep learning enables emergence of features automatically without explicit feature engineering, foundation models enable emergence of new functionalities that help them to excel in multiple downstream tasks.

Homogenisation is the consolidation of techniques for building ML systems that can effectively work for different tasks. Deep learning is built around the homogenisation of architectures such as CNNs, RNNs, etc. Be it any task, a deep learning model will typically consist of known neural network architectures involving a different combination of RNNs, CNNs, pooling, Softmax layers, etc. On the other hand, foundation models go one step further and are built around the concept of ‘homogenisation of models’ themselves. As we have seen, almost all pretrained models or large language models are built on top of transformer models. With their self-attention layers these models help to take advantage of hardware parallelism, and can scale even with a large number of parameters and very long sequences.

Homogenisation of models has moved across modalities, namely, language, vision, speech and multimodal. While this allows the cross-leverage of improvements, it also has the disadvantage that any problem associated with the foundation model will now have a wider impact on AI deployment. For example, many of the pretrained language models have been identified to have issues due to ‘social biases’ in the data they have trained on, leading them to discriminate based on these biases. The models may also amplify these biases and toxicity present in their training data.

Because of the vast scales of data on which they have been trained, the emergence of new functionalities has been associated with foundation models. For example, a foundation model like GPT-3 can be adapted to a new task by suitable prompting or in-context learning, wherein the model is provided with a natural language description of the task and is able to carry it out reasonably well. Since ‘emergence of new functionalities’ is not well understood, as the model has not been explicitly trained on data for that task, the model behaviour can be difficult to understand, explain and can also have unexpected failure modes which come to light only after deployment in the field. Thus ‘emergence’ is both a boon and bane associated with foundation models. The key question is whether ‘emergence of new functionalities’ can be understood and explained by an auxiliary mechanism, with failure modes anticipated and quantified. This is still an open research problem.

While foundation models have their humble origin in language, their potential is definitely in multi-modal data learning. For instance, much of common sense and world knowledge is not explicitly specified through text. To give a trivial example, our ‘social norms’ tell us that it is fine to stare at the sunset on a beach, but not to stare at a stranger on a train. Such norms and common sense knowledge are not codified in data, and hence pretrained models may not be able to learn them if they are trained only on text. However, by training them on multi-modal data such as real world images, videos and text, models can implicitly learn common sense knowledge and social norms. This will also allow them to learn embodied physical knowledge. Over the next couple of columns, we will be discussing the capabilities and challenges of foundation models, and how this promises to be an exciting area of research in artificial intelligence in the near future.

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